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Machine Learning

ML101 | Suggested sequence · Semester 3 | Student-authored study guide

How to use these notes

Supervised and unsupervised learning, feature preparation, validation, metrics, and model limitations. Work through the units in order, reproduce each example, then answer the review prompts from memory. This is supplemental practice material, not an official MBU syllabus or assessment. Course sequencing is suggested and should be checked against your current approved syllabus.

MODULE 01 · Problem framing and baselines

Choose supervised learning when labeled targets exist; use unsupervised methods to explore structure without a target. Establish a simple baseline and split data before fitting or selecting transformations.

Worked example

```
Predicting a numeric energy use is regression; predicting one of three equipment classes is classification.
```

Check your understanding

- What is a target variable?
- Why establish a baseline first?

MODULE 02 · Pipelines and evaluation

Fit preprocessing only on training data and apply the learned transformation to validation/test data. Cross-validation estimates variability across folds. Keep a final test set for an unbiased final estimate.

Worked example

```
Use a pipeline that imputes missing values and scales features within each training fold.
```

Check your understanding

- How can scaling leak information?
- When is stratification useful?

MODULE 03 · Metrics, fairness, and monitoring

Use MAE/RMSE for regression and precision/recall/F1 or ROC-AUC for classification as appropriate. Examine subgroup errors, data drift, and limitations. A model score does not justify a high-impact decision by itself.

Worked example

```
When missing a positive case is costly, inspect recall and threshold behavior rather than maximizing accuracy alone.
```

Check your understanding

- What trade-off changes with a classification threshold?
- What is model drift?

Revision checklist

- %¡ Can I explain the main idea without copying the notes?
- %¡ Can I reproduce the example and state its assumptions?
- %¡ Can I test an edge case and describe what should happen?
- %¡ Have I recorded a limitation and checked a primary reference?

Further reading

Use the official documentation and open references linked in the portal. Check the publication/version and license at the source before redistribution. Cite the source and access date in coursework.